

Thenmozhi Boopathy

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PROFESSIONAL SUMMARY

Career-changer from a math-teaching background applying that rigor to applied GenAI engineering — built and deployed an agentic RAG chatbot on Databricks (LangChain, Spark, Delta Lake, MLflow), with coursework spanning deep learning architectures for NLP and GenAI.

EDUCATION

Miami University <i>Master of Science in Business Analytics</i> U.S. Permanent Resident (Green Card) - no visa sponsorship required	Oxford, OH Expected Graduation 2027
Bharathidasan University <i>Master of Philosophy in Mathematics</i> <i>Master of Science in Mathematics</i> <i>Bachelor of Science in Mathematics</i>	TamilNadu, India 2008 2007 2005

TECHNICAL SKILLS

Certification: Databricks University Alliance Student — Databricks (issued 2026). TNSET (Tamil Nadu State Eligibility Test) - 2017
ML / AI: Pandas, ML Pipeline, LangChain, MLflow, Vector Search, GenAI / LLM, Agentic AI, Deep Learning, NLTK
Data Platforms: Databricks, Apache Spark, Spark MLlib, Delta Lake, NoSQL (MongoDB), Power BI
Languages & Tools: Python, R, SQL, Jupyter Notebook, RStudio, Visual Studio Code, GIT

RELEVANT COURSEWORK

Big Data: Core Python, Pandas, NoSQL (MongoDB), Web APIs, Scikit-learn, Spark MLlib, Text Mining (NLTK, TF-IDF, Word2Vec).
Cloud & GenAI: Databricks, Apache Spark, Delta Lake, Unity Catalog, Databricks Vector Search, LangChain, MLflow, Agentic AI, Recommender Systems, Retrieval-Augmented Generation (RAG).
Data Mining: Exploratory Data Analysis, Dimension Reduction, Tree-Based Models, Logistic Regression, Neural Networks, Model Evaluation,
Current Coursework (Fall 2026, in progress): Power BI & DAX, Data Storytelling & Dashboards, Linear/Integer Programming (Gurobi), Reproducible Analytics & Business Value Proposition.

PROJECT EXPERIENCE

Enterprise RAG Chatbot — Job Finder Assistant (ISA-632) Built and deployed an enterprise-grade Retrieval-Augmented Generation (RAG) chatbot on Databricks with a 3-person team: used Apache Spark to ingest 2,500+ job records into Delta Tables, integrated Databricks Vector Search with a LangChain agent on Llama 4 Maverick, and built a Streamlit UI with MLflow-tracked evaluation, achieving 89% positive user feedback.	2026
End-to-End R Machine Learning Workflow for Loan Default Prediction (ISA-591) Built an end-to-end R pipeline (decision trees, logistic regression, neural network, XGBoost) to predict loan default on an imbalanced dataset (80.8% no-default / 19.2% default); diagnosed class imbalance behind a 0.069 baseline sensitivity, then raised it to 0.64 (AUC 0.74) via downsampling and cross-validated pruning, and tuned the decision threshold with Youden's J to minimize costly false negatives.	2025
Theme Park Attendance Forecasting Pipeline with Python (ISA-514) Built a Python-based machine learning pipeline using MongoDB for data storage and integration of structured and unstructured sources (weather, wait times, schedules, Reddit sentiment) to forecast Six Flags attendance for operational decision-making.	2025

PROFESSIONAL EXPERIENCE

S.A College of Engineering <i>Assistant Professor</i> ● Delivered instruction in quantitative and statistical methods (numerical analysis, probability, matrices, and applied functions), emphasizing analytical problem-solving skills relevant to data-driven business decision-making.	Chennai, India Jan–May 2015
Sri Venkateswara College of Engineering <i>Assistant Professor</i> ● Performed lecturing in advanced quantitative methods (calculus, sequences, integrals), building strong analytical and problem-solving foundations relevant to data modeling and forecasting.	Chennai, India 2012-2013
Bon Secours College for Women <i>Lecturer (PG and UG)</i> ● Conducted training in mathematics and analytics concepts for undergraduate and postgraduate students, strengthening quantitative reasoning, data interpretation, and problem-solving skills.	Tanjore, India 2009-2012

VOLUNTEER ACTIVITIES

Language Teacher <i>Greater Cincinnati Tamil School, Cincinnati</i> ● Designed and provided customized language programs for students of varying proficiency levels, using assessments to track progress and enhance outcomes. Coordinated program logistics and engaged stakeholders to promote the Seal of Biliteracy.	Cincinnati, Ohio 2021-2025
Gifted Education Co-Representative <i>Sycamore Gear</i> ● Designed and executed student engagement initiatives, enhancing community involvement and program effectiveness.	Cincinnati, Ohio 2023-2024